

Project Charters, Scope, Assumptions, and Constraints

Lab

DAEN 459 - Fall 2026
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A project charter for a capstone research project is a short agreement about what the project will investigate, why it matters, what the team will deliver, and how progress will be judged.

Aim for 2–4 pages. It should be specific enough to guide decisions, while leaving room for the research findings to shape the final work.

Project charter: sections and guiding questions

Section	What to include	Guiding question
Project overview	Title, team, advisor, and dates	What is this project?
Problem and significance	Problem, affected people, and importance	Why does it matter?
Research question	Primary and supporting questions	What will we answer?
Objectives	Concrete research outcomes	What will we accomplish?
Scope	Population, data, setting, and boundaries	What is included?
Approach	Sources, methods, and approvals	How will we investigate?
Deliverables	Report, code, presentation, or other outputs	What will we submit?
Milestones	Key tasks and target dates	When is each stage due?
Success criteria	Standards for completion and quality	How will we assess the work?
Risks and dependencies	Obstacles and fallback plans	What could delay us?
Roles and approval	Responsibilities and decision authority	Who approves changes?

What the charter does

- Records the question, scope, approach, outputs, and decisions that guide the project.
- Gives the student and advisor a shared reference when the project changes.
- Usually fits in two to four pages. Write it before extensive data collection, then revise with approval.

Drafting task

Start with a one-sentence description: “This project investigates ... in ... using”

1. Project overview

- Give the project a specific working title that names the subject and setting.
- List the student or team, advisor, partner if any, and expected project dates.
- Write a two-sentence summary that someone outside the course could understand.

Drafting task

Complete: “We will study [subject] in [setting] to understand [purpose].”

Check

Could a reader identify the project without seeing the rest of the charter?

2. Problem and significance

- Describe the current problem with a specific observation or source.
- Name who experiences the problem or who would use the findings.
- Explain what remains unknown and why an answer would matter.

Drafting task

Write one paragraph: situation, evidence of the problem, and the knowledge gap. Cite any factual claims.

Check

Does the paragraph explain why research is needed, beyond saying the topic is interesting?

3. Research question

- State one primary question that the available time and data can answer.
- Specify the population, setting, variables, or process where relevant.
- Add supporting questions only when each contributes to the primary answer.

Drafting task

Try: “How does [factor or method] relate to [outcome] in [setting]?” Adjust the wording to fit your design.

Check

Can you name the evidence that would answer this question?

4. Objectives

- Describe what you will do to answer the question: compare, estimate, explain, or evaluate.
- Make each objective observable in the final report.
- Keep objectives distinct from hoped-for results.

Drafting task

Write two or three numbered objectives, each starting with an action verb and naming its output.

Check

Could a reader determine whether each objective was completed?

5. Scope and boundaries

- Identify the population, time period, location, data, and methods you will include.
- State exclusions that a reader might reasonably expect you to cover.
- Explain any practical limit that affects how widely findings can apply.

Drafting task

Fill in: “This project includes [specific work] and excludes [related work].”

Check

Would this section help settle a request to add a new dataset or analysis midway through the term?

6. Research approach

- Name the data sources and how you will obtain or collect them.
- Outline the analysis and why it fits the question.
- Identify permissions, ethics review, privacy measures, or tools needed before work begins.

Drafting task

Sketch the sequence: obtain evidence, prepare it, analyze it, and interpret the results.

Check

Is each planned method feasible with the access and time you actually have?

7. Deliverables

- List each item to be submitted: proposal, report, presentation, code, or other agreed output.
- State its audience and expected form or length where useful.
- Distinguish a research output from a possible finding.

Drafting task

For each item, write “[Deliverable] for [audience], containing [essential components].”

Check

Can the advisor verify each item at submission time?

8. Milestones

- Work backward from the final presentation and submission dates.
- Include approval, data access, collection, analysis, draft, feedback, and revision.
- Allow time for delays where work depends on other people or systems.

Drafting task

Make a table with milestone, target date, and evidence of completion.

Check

Does the schedule leave time to revise after feedback?

9. Success criteria

- State what makes the work complete and credible, regardless of the result.
- Include relevant standards for data quality, method, reproducibility, and communication.
- Tie criteria to the objectives and required deliverables.

Drafting task

Write criteria that can be checked, such as documented cleaning decisions and a reproducible analysis.

Check

Would the project still succeed if the hypothesis is unsupported?

10. Risks, roles, and decisions

- Identify the most likely obstacles: access, recruitment, approvals, missing data, or time.
- Pair each major risk with a practical fallback.
- Assign responsibilities and say who approves changes to scope or method.

Drafting task

Use three columns: risk or decision, responsible person, and response.

Check

If a critical dataset becomes unavailable, does the charter say what happens next?

Putting the charter together

One-paragraph synthesis

This project will investigate [question] in [setting] using [data and method]. It will produce [deliverables] by [date]. Its scope is [boundary], and completion will be assessed using [criteria].

1. Draft each section in plain language.
2. Check that the question, method, scope, and deliverables agree.
3. Review the draft with the advisor and record approved changes.

Example: data quality capstone

Question: How do revised validation rules affect completeness and reliability in a university research dataset?

- **Scope:** One named dataset and a defined reporting period.
- **Method:** Apply both rule sets and compare specified quality measures.
- **Outputs:** Reproducible code, documented results, final report, and presentation.
- **Risk:** If access is delayed, use an approved sample or revise the schedule with the advisor.

Check

The charter commits to evaluating the rules, without assuming they will improve quality.